VTOL avionics carrier rev 3 (ESP-12F, 1S logic battery + on-board 5 V boost converter U8 with under-voltage lock-out U9; assembled by JLCPCB except U1-U4 and the sensor modules), 76 x 7

Nets connect by global labels (net names = spec nets). Logic GND and GND_PWR meet only at NT1 (Power sheet).

NOT_VERIFIED: never built or powered. Module pin orders (GY-521, BMP280, OpenLog, NEO-7M lead, 1S pack plug) are assumptions - check silkscreen. U8 uses a generic 6-pin symbol: ERC gives

Power

: power.kicad_sch

NAV

: nav.kicad_sch

FCU

: fcu.kicad_sch

COM

: com.kicad_sch

GNS

: gns.kicad_sch

Level_shift

: level_shift.kicad_sch

IO_connectors

: io.kicad_sch

NOT_VERIFIED: never built or powered
generated by scripts/build_schematic.py (hierarchy) + ai_eda SchematicCompiler (sheets)

Sheet: /
File: vtol_carrier.kicad_sch

Title: VTOL avionics carrier

Size: A4	Date:	**Rev: 3**
KiCad E.D.A. 10.0.6		Id: 1/8

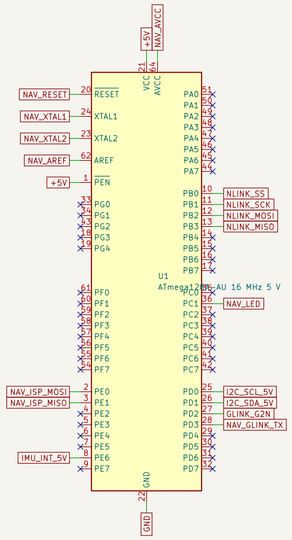
C41
22u 10V
C42
22u 10V
C43
22u
C44
10u
C45
22u 10V tantalum (TAJA226K010RNJ, +-10 %)
C46
100n
C57
22u
C58
1n 50V X7R
C59
100n
C60
100n
J14
Molex SPOX 5267-02A 2.50 mm 2-pin vertical, mates with JST-PH S2B-PH-K-S
J15
JST-PH S2B-PH-K-S 2-pin vertical, mates with JST-PH B2B-PH-K-S
L5
Coilcraft FXL0420-1R0-MA1
Q10
AO3401A (−30 V, −4 A, Vgs(th) −0.5…−1.3 V, Rds(on) 65 mOhm at Vgs −2.5 V; A)
R14
65kOhm
R15
47k 1%
R16
47k 1%
R63
150k 1%
R64
20k 1%
R65
470k 1%
R66
270k 1%
R67
3.3M 1%
R68
1M 1%
U5
AMS1117-3.3
U6
TPS61023DRLR
U9
TLV3012AIDBVR
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG
PWR_FLAG

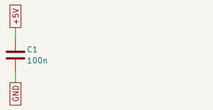
NOT_VERIFIED: never built or powered
generated by scripts/build_schematic.py via ai_eda SchematicCompiler from spec/carrier_spec.json

Sheet: /Power/
File: power.kicad_sch

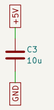
Title: VTOL carrier — 1S logic battery input, reverse-polarity P-FET, power-s

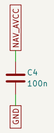
Size: A2	Date:	Rev: 3
KiCad E.D.A. 10.0.6		Id: 2/8

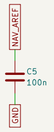










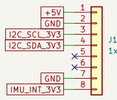


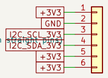


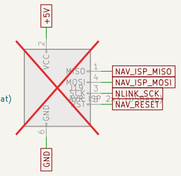


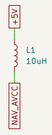
















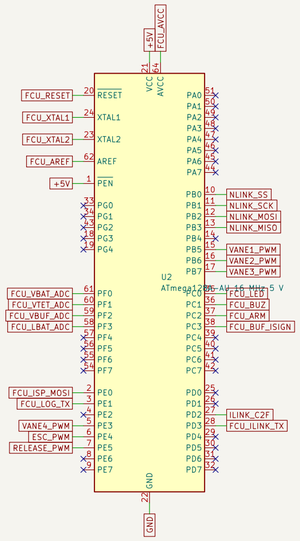


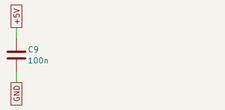
NOT_VERIFIED: never built or powered
generated by scripts/build_schematic.py via ai_eda SchematicCompiler from spec/carrier_spec.json

Sheet: /NAV/
File: nav.kicad_sch

Title: VTOL carrier — NAV ATmega128A + GY-521 IMU + BMP280 barometer

Size: A2	Date:	Rev: 3
KiCad E.D.A. 10.0.6		Id: 3/8



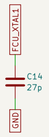


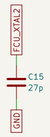






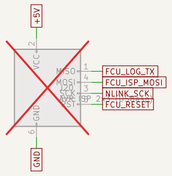




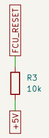


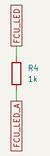


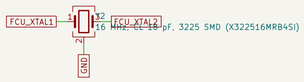










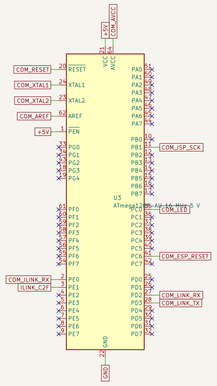


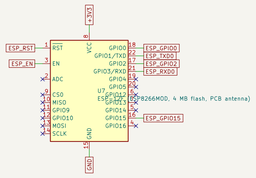
NOT_VERIFIED: never built or powered
generated by scripts/build_schematic.py via ai_eda SchematicCompiler from spec/carrier_spec.json

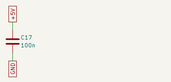
Sheet: /FCU/
File: fcu.kicad_sch

Title: VTOL carrier — FCU ATmega128A

Size: A2	Date:	Rev: 3
KiCad E.D.A. 10.0.6		Id: 4/8

























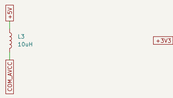


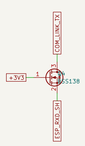


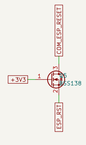












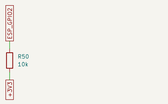
















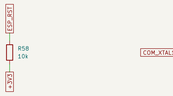








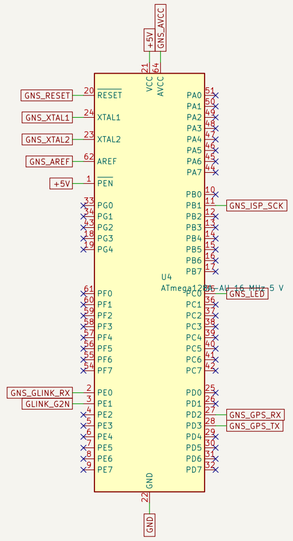


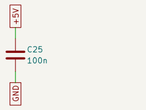




NOT_VERIFIED: never built or powered
generated by scripts/build_schematic.py via ai_eda SchematicCompiler from spec/carrier_spec.json

Sheet: /COM/	
File: com.kicad_sch	
Title: VTOL carrier — COM ATmega128A + ESP-12F Wi-Fi bridge (soldered),	
Size: A1 Date:	**Rev: 3**
KiCad E.D.A. 10.0.6	Id: 5/8







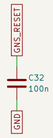






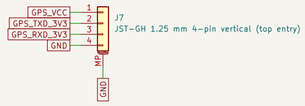


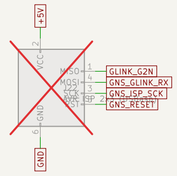


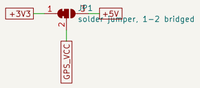




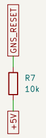




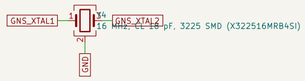










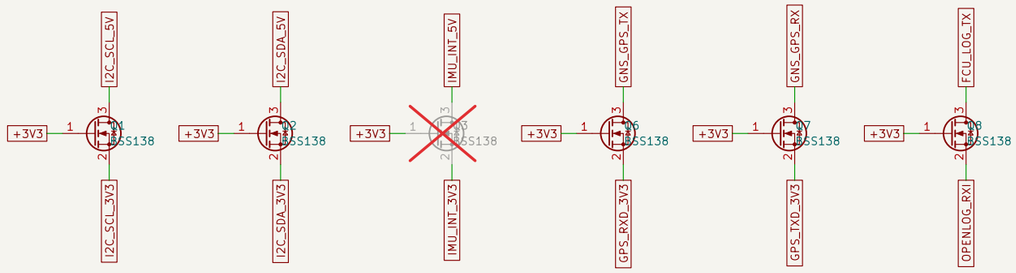


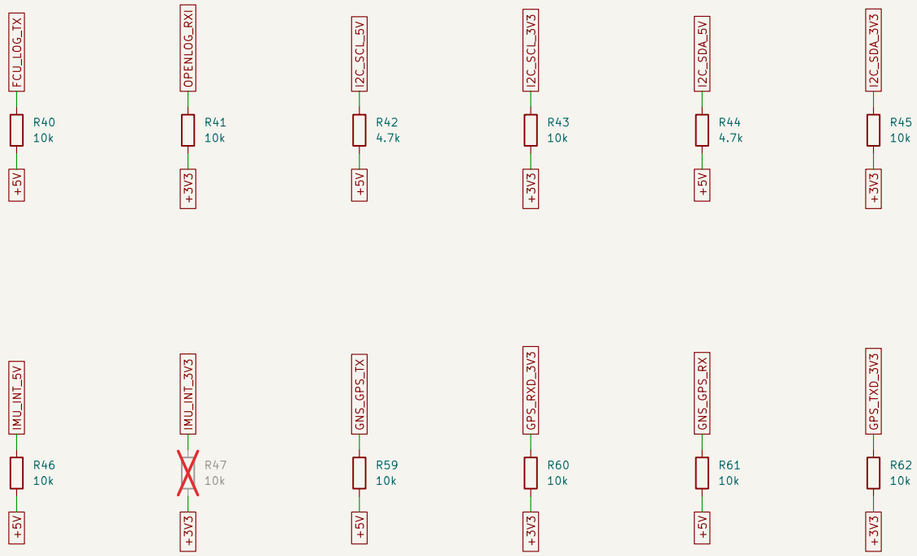
NOT_VERIFIED: never built or powered
generated by scripts/build_schematic.py via ai_eda SchematicCompiler from spec/carrier_spec.json

Sheet: /GNS/
File: gns.kicad_sch

Title: VTOL carrier – GNS ATmega128A + NEO-7M GPS lead

Size: A2	Date:	Rev: 3
KiCad E.D.A. 10.0.6		Id: 6/8



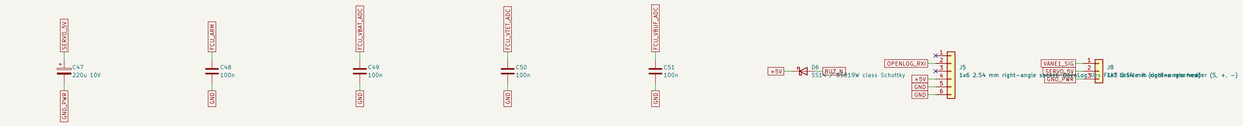


NOT_VERIFIED: never built or powered
generated by scripts/build_schematic.py via ai_eda SchematicCompiler from spec/carrier_spec.json

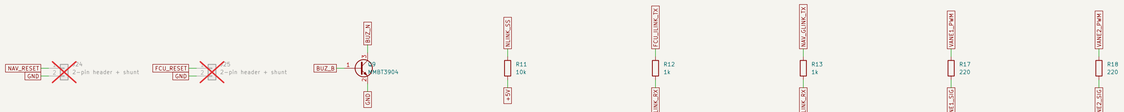
Sheet: /Level_shift/
File: level_shift.kicad_sch

Title: VTOL carrier – BSS138 level shifters 5 V <–> 3.3 V

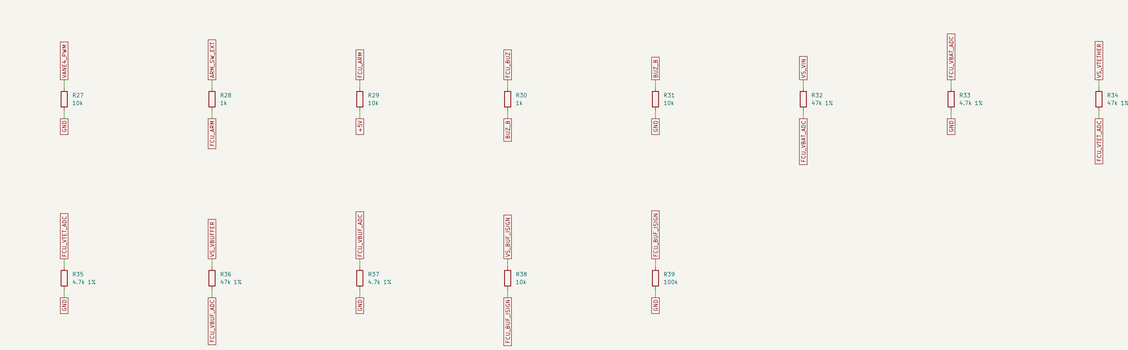
Size: A3	Date:	**Rev: 3**
KiCad E.D.A. 10.0.6		Id: 7/8











NOT_VERIFIED: never built or powered
generated by scripts/build_schematic.py via ai_eda SchematicCompiler from spec/carrier_spec.json

Sheet: /IO_connectors/
File: io.kicad_sch

Title: VTOL carrier — Servo / ESC / arm / buzzer / power-sense headers, I

Size: A1	Date:	Rev: 3
KiCad E.D.A. 10.0.6		Id: 8/8